

## Lab 05: LOOP STATEMENTS I

### Objective

The purpose of this Lab. is to familiarize student with how to write programs that repeatedly execute one or more statements with the use of repetition control structure **while**.

### Current Lab Learning Outcomes (LLO)

Student will learn how to write Java programs using loop statement, **while**, to repeat Java statements more than times. Students will practice several kinds of problems using loop statement, **while**. There are some exercises, through which they will understand the concept learn in this chapter.

### Lab Requirements

Net Beans IDE 8

### Lab Description

- Java provides a powerful construct called a loop. A loop controls how many times an operation (or a sequence of operations) is performed.
- Java provides three types of loop statements (**while** loops, **do-while** loops and **for** loops).

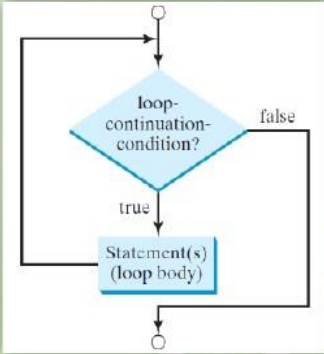
### The **while** Loop

☞ A **while** loop executes statements repeatedly while the condition is true

☞ Syntax:

```
while (loop-continuation-condition) {  
    // loop body  
    Statement(s);  
}
```

- the "loop body" is the part that contains the repeated statements
- a one-time execution of the loop body is called an **iteration**
  - ◆ an iteration of the loop



**Example:** Print "Welcome to Java" one hundred times using while statement

```
int count = 0;
while (count < 100)
    System.out.println("Welcome to Java!");
    count++;
}
```

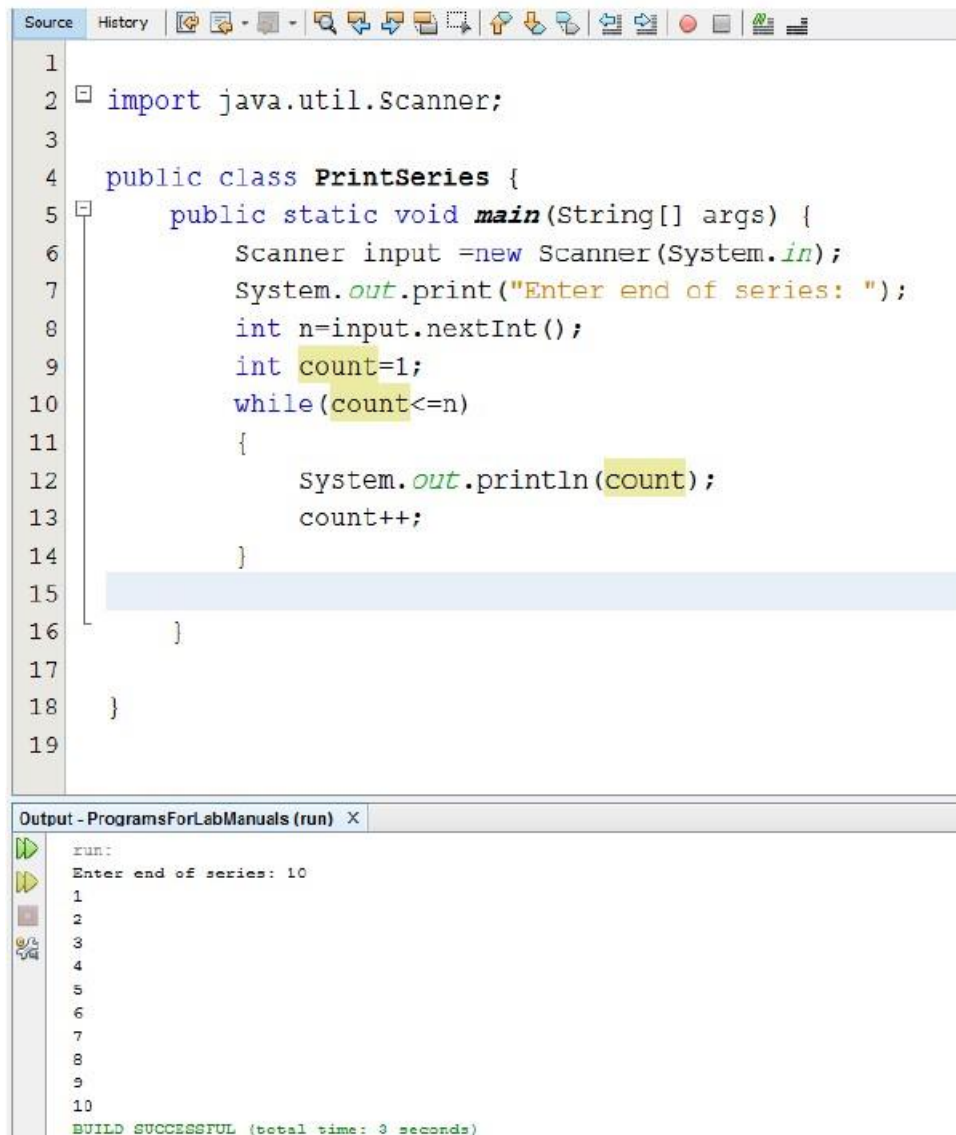
## Lab Assessment

**Practice Activity with Lab Instructor** (10 minutes)

**(Printing Series)** Write a program that displays numbers for **1** to a given number, **n**.

**Solution:**

1. Open NetBeans and create a new project
2. Create a new java main class and write its name as **PrintSeries**
3. Write the following code inside the main method



```
Source History
1
2 import java.util.Scanner;
3
4 public class PrintSeries {
5     public static void main(String[] args) {
6         Scanner input = new Scanner(System.in);
7         System.out.print("Enter end of series: ");
8         int n = input.nextInt();
9         int count = 1;
10        while(count <= n)
11        {
12            System.out.println(count);
13            count++;
14        }
15    }
16 }
17
18 }
19
```

Output - ProgramsForLabManuals (run) X

```
Run:
Enter end of series: 10
1
2
3
4
5
6
7
8
9
10
BUILD SUCCESSFUL (total time: 3 seconds)
```

**Short Exercise (5 minutes)**

Rewrite the above program such that to display the even numbers.

**Individual Activities: (60 minutes)**

1. **(Conversion from kilograms to pounds)** Write a program that displays the following table (note that 1 kilogram is 2.2 pounds):

| Kilograms | Pounds |
|-----------|--------|
| 1         | 2.2    |
| 3         | 6.6    |
| ...       |        |
| 197       | 433.4  |
| 199       | 437.8  |

2. **(Print first n positive numbers)** Write a program to prompt the user to enter one integer, **n**. Then you should print the first **n** positive numbers that are divisible by 7, one number per line. So if the user enters the value 5 (so  $n = 5$ ), you should print the first five positive integers that are divisible by 7, which are 7, 14, 21, 28, and 35.
3. **(Count positive and negative numbers and compute the average of numbers)** Write a program that reads an unspecified number of integers, determines how many positive and negative values have been read, and computes the total and average of the input values (not counting zeros). Your program ends with the input 0. Display the average as a floating-point number. **Here is a sample run:**

```
Enter an integer, the input ends if it is 0: 1 2 -1 3 0 Enter
The number of positives is 3
The number of negatives is 1
The total is 5.0
The average is 1.25
```

```
Enter an integer, the input ends if it is 0: 0 Enter
No numbers are entered except 0
```

4. **(Financial application: compute future tuition)** Suppose that the tuition for a university is \$10,000 this year and increases 5% every year. In one year, the tuition will be \$10,500. Write a program that computes the tuition in ten years and the total cost of four years' worth of tuition after the tenth year.